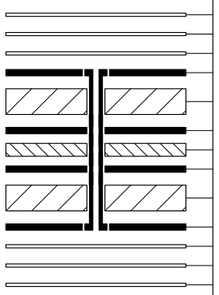
Heartrate-DevKit-HW Fabrication Document

Layer Stack Legend



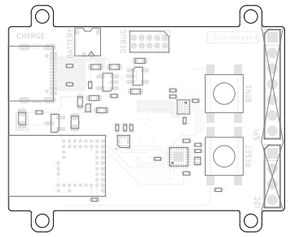
Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen			Legend	GBR
	F.Mask	0.01mm		Solder Mask	GBR
Copper	**F.Cu**	**0.035mm (1oz)**		**Signal**	**GBR**
Core		0.196mm	FR4	Dielectric	
Copper	**Inner1**	**0.035mm (1oz)**		**Plane**	**GBR**
Prepreg		0.43mm	FR4	Dielectric	
Copper	**Inner2**	**0.035mm (1oz)**		**Plane**	**GBR**
Core		0.196mm	FR4	Dielectric	
Copper	**B.Cu**	**0.035mm (1oz)**		**Signal**	**GBR**
	B.Mask	0.01mm		Solder Mask	GBR
	B.Silkscreen			Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 0.982mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [%]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers

Top Fabrication (Scale 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

4) SURFACE FINISH: HAL SNPB

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING BLACK NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 150 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY ZSWATCH.

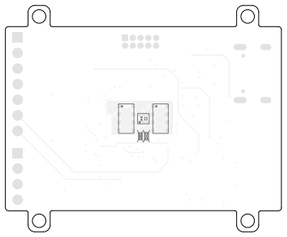
10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	45.000 × 36.900 mm
BOARD THICKNESS	0.982 mm
TRACE WIDTH	0.150 mm
TRACE TO TRACE	0.100 mm
MIN. HOLE (PTH)	0.150 mm
MIN. HOLE (NPTH)	0.650 mm
ANNULAR RING	0.100 mm
COPPER TO HOLE	0.100 mm
COPPER TO EDGE	0.100 mm
HOLE TO HOLE	0.100 mm

Comments: ZSWatch	Company: **ZSWatch**		Variant: CHECKED	Git Hash: 10ac9d5
	Board Name: **Heartrate-DevKit-HW**		Project Name: **ZSWatch Heartrate-DevKit**	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: ZSWatch-Heartrate-DevKit.kicad_pcb	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **1** of **10**

Heartrate-DevKit-HW Fabrication Document

Bottom Fabrication (Scale 1:1)



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Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: ZSWatch-Heartrate-DevKit.kicad_pcb	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **2** of **10**

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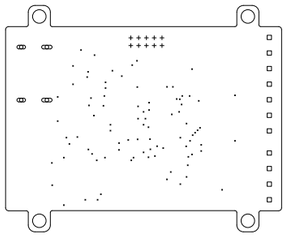
Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	88	0,15mm (5,91mils)	PTH	Round	F.Cu - B.Cu	Via
○	4	0,60mm (23,62mils)	PTH	Slot	F.Cu - B.Cu	Pad
+	10	0,65mm (25,59mils)	PTH	Round	F.Cu - B.Cu	Pad
□	11	1,00mm (39,37mils)	PTH	Round	F.Cu - B.Cu	Pad
	Total 113					



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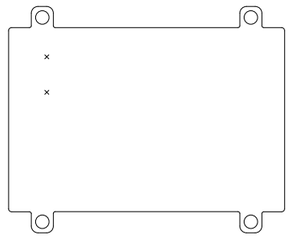
	Board Name: **Heartrate-DevKit-HW**		Project Name: **ZSWatch Heartrate-DevKit**	
Sheet Title: Drill Drawing (L1 - L4)	File Name: ZSWatch-Heartrate-DevKit.kicad_pcb	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **3** of **10**

Heartrate-DevKit-HW Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	2	0,65mm (25,59mils)	NPTH	Round	F.Cu - B.Cu	Mechanical
	Total 2					



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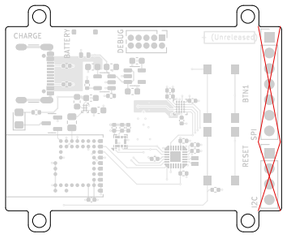
	Board Name: **Heartrate-DevKit-HW**		Project Name: **ZSWatch Heartrate-DevKit**	
Sheet Title: Drill Drawing (L1 - L4)	File Name: ZSWatch-Heartrate-DevKit.kicad_pcb	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **4** of **10**

Heartrate-DevKit-HW Fabrication Document

Top Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]

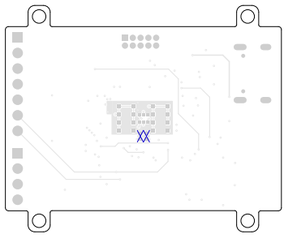
Ref.	Net	X [mm]	Y [mm]



Comments: ZSWatch	Company: **ZSWatch**		Variant: CHECKED	Git Hash: 10ac9d5
	Board Name: **Heartrate-DevKit-HW**		Project Name: **ZSWatch Heartrate-DevKit**	
Sheet Title: Top Test Points (Scale 1:1)	File Name: ZSWatch-Heartrate-DevKit.kicad_pcb	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **5** of **10**

Heartrate-DevKit-HW Fabrication Document

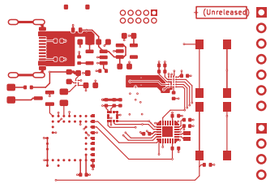
Bottom Test Points (Scale 1:1)



Comments: ZSWatch	Company: **ZSWatch**		Variant: CHECKED	Git Hash: 10ac9d5
	Board Name: **Heartrate-DevKit-HW**		Project Name: **ZSWatch Heartrate-DevKit**	
Sheet Title: Bottom Test Points (Scale 1:1)	File Name: ZSWatch-Heartrate-DevKit.kicad_pcb	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **6** of **10**

Heartrate-DevKit-HW Fabrication Document

F.Cu (Scale 1:1)



Comments: ZSWatch	Company: **ZSWatch**		Variant: CHECKED	Git Hash: 10ac9d5
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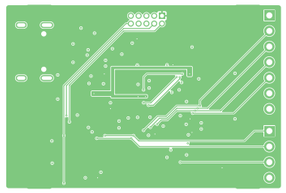
Sheet Title: F.Cu (Scale 1:1)	File Name: ZSWatch-Heartrate-DevKit.kicad_pcb	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **7** of **10**

Heartrate-DevKit-HW Fabrication Document

Inner1 (Scale 1:1)



Comments: ZSWatch	Company: **ZSWatch**		Variant: CHECKED	Git Hash: 10ac9d5
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Sheet Title: Inner1 (Scale 1:1)	File Name: ZSWatch-Heartrate-DevKit.kicad_pcb	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **8** of **10**

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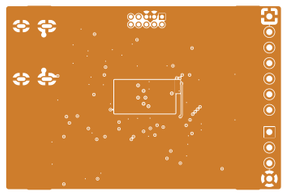
Inner2 (Scale 1:1)



Comments: ZSWatch	Company: **ZSWatch**		Variant: CHECKED	Git Hash: 10ac9d5
	Board Name: **Heartrate-DevKit-HW**		Project Name: **ZSWatch Heartrate-DevKit**	
Sheet Title: Inner2 (Scale 1:1)	File Name: ZSWatch-Heartrate-DevKit.kicad_pcb	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **9** of **10**

Heartrate-DevKit-HW Fabrication Document

B.Cu (Scale 1:1)

Comments: ZSWatch	Company: **ZSWatch**		Variant: CHECKED	Git Hash: 10ac9d5
	Board Name: **Heartrate-DevKit-HW**		Project Name: **ZSWatch Heartrate-DevKit**	
Sheet Title: B.Cu (Scale 1:1)	File Name: ZSWatch-Heartrate-DevKit.kicad_pcb	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **10** of **10**